Akoizo Original Practice - Form 5 - Answer key

Division C | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Quadrupling the canopy area changes terminal velocity by a factor of:

A. 1/2

B. 1/4

C. 2

D. 1/16

Correct: A

2. (2 points) For stable flight, the center of gravity should be:

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

A. Ahead of the center of pressure

B. Behind the center of pressure

C. At the same point as the center of pressure

D. At the nozzle

Correct: A

3. (2 points) A team has an incomplete practice log and a clean 16.00 s launch. What is its score?

A. 13.60

B. 16.00

C. 11.20

D. 4.00

Correct: A

4. (2 points) Why is release near apogee ideal?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

A. It is required by the rules

B. It avoids the 0.85 multiplier

C. The parachute begins its descent from the greatest height at low speed

D. It makes the rocket fly higher

Correct: C

5. (4 points) Calculate apogee for an observer 12 m away sighting 40° (tan 40° about 0.84), eye height 1.6 m.

answer: 12 x 0.84 = 10.1 m above eye, plus 1.6 m = about 11.7 m (about 38 ft). (4 points).

6. (4 points) Why does a very light rocket not always fly higher?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

answer: It accelerates quickly but has little momentum, so drag slows it faster, and it may be less stable in flight. (4 points).

7. (4 points) Write a one-sentence pressure justification for a 22-foot ceiling, using example data.

answer: In seven test flights at 55 psi our rocket peaked between 17 and 19 feet, so 55 psi clears the 22-foot ceiling by at least 3 feet. (4 points).

8. (4 points) Why does adding tape to the payload matter?

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

answer: Tape adds weight, which raises terminal velocity and shortens descent; use only what holds reliably. (4 points).

9. (6 points) Solve this case and explain your reasoning: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?

reasoning: Stability margin = 21 - 18 = 3 cm.
In body diameters: 3 ÷ 8 about 0.4.
Below the 1-diameter guideline: enlarge the fins or add nose mass. (3 points).

conclusion: The rocket is barely stable; moving the CG forward or the CP back improves straight flight. (3 points).

10. (6 points) Solve this case and explain your reasoning: A team with a complete log records 14.20 s on launch 1 (clean) and 19.80 s on launch 2, but launch 2 grazed a light on the way up. What is its score?

reasoning: Launch 1: 14.20 s x 1.00 = 14.20.
Launch 2: timing stopped at contact; even 19.80 s x 0.25 = 4.95.
Best launch: 14.20. (3 points).

conclusion: The clean, shorter flight wins easily: avoiding the ceiling matters more than climbing higher. (3 points).

11. (6 points) Solve this case and explain your reasoning: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?

reasoning: Early release means the payload sits too loosely or is pushed out by airflow.
Make the nose cup slightly deeper or snugger so it stays through the fast climb.
Retest with video until release happens near apogee. (3 points).

conclusion: Adjust the fit so separation happens as the rocket slows near the top, not during the fast climb. (3 points).

12. (2 points) Which launch pressure is best justified?

Supplied synthetic observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

A. About 50 psi

B. 65 psi

C. 45 psi

Correct: A

13. (2 points) Define "Thrust" in the context of Ping Pong Parachute.

answer: Force from air leaving the nozzle. (2 points).

14. (2 points) Define "Required parameters" in the context of Ping Pong Parachute.

answer: Pressure (psi), peak height (ft), and time aloft (s). (2 points).

15. (2 points) Define “Drag coefficient (C_d)” in the context of Ping Pong Parachute.

answer: Number describing how much drag a shape makes. (2 points).

16. (2 points) Define “Ceiling height” in the context of Ping Pong Parachute.

answer: Room height given to teams in advance, in feet. (2 points).

17. (2 points) Define “Drag” in the context of Ping Pong Parachute.

answer: Air resistance, growing with the square of speed. (2 points).

18. (2 points) Define “Outlier” in the context of Ping Pong Parachute.

answer: A result far from the rest, which should be explained. (2 points).